

Portfolio.

Amaljith P A

MECHANICAL ENGINEERING

Mechanical Design

CAD · Product
Development

Hello. I'm Amaljith

Mechanical Engineer with a passion for designing practical, efficient and user-focused products that solve real-world problems.

ABOUT ME

I am a Mechanical Engineering graduate who enjoys turning ideas into functional solutions. My work combines design thinking, engineering fundamentals and hands-on execution.

I have experience in CAD design, component development, fabrication, assembly and testing through academic projects and self-initiated work.

I am constantly learning new tools and technologies to improve product performance and create meaningful impact.

CORE AREAS



Mechanical Design

3D modelling, part design and detailed engineering drawings.



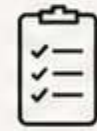
CAD

Proficient in SolidWorks and AutoCAD for product development.



Fabrication & Assembly

Hands-on experience in fabrication, assembly and fitment of components.



Product Development

From concept and design to prototyping and testing.

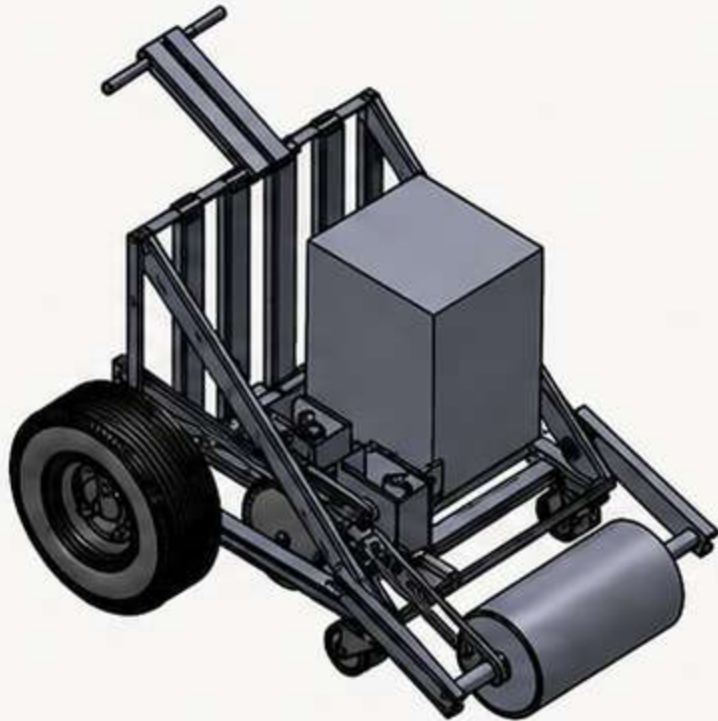
FEATURED PROJECT

Compact Pothole Repair Machine.

A semi-automatic machine designed to repair potholes quickly and efficiently. It integrates cutting, material filling and compaction in a compact, mobile setup to improve road maintenance and reduce manual effort.



FABRICATED PROTOTYPE



3D CAD ASSEMBLY



PROJECT TYPE
Academic Project



DOMAIN
Mechanical Design
& Fabrication

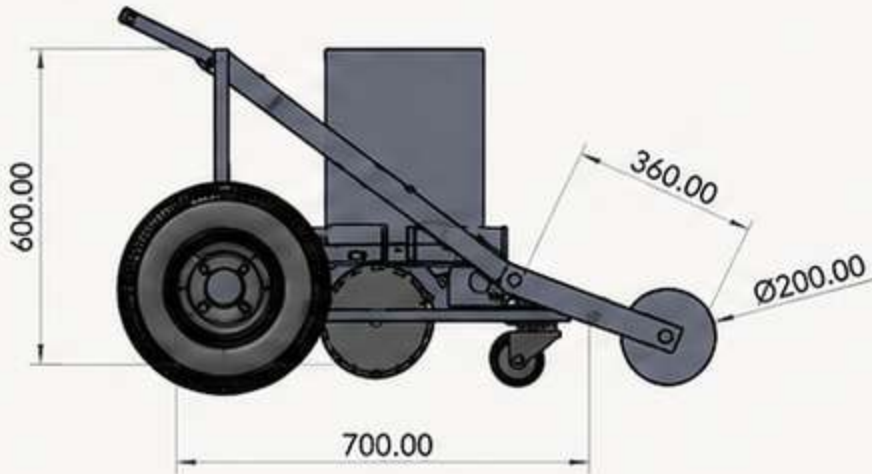


FOCUS
Design · Fabrication
· Performance

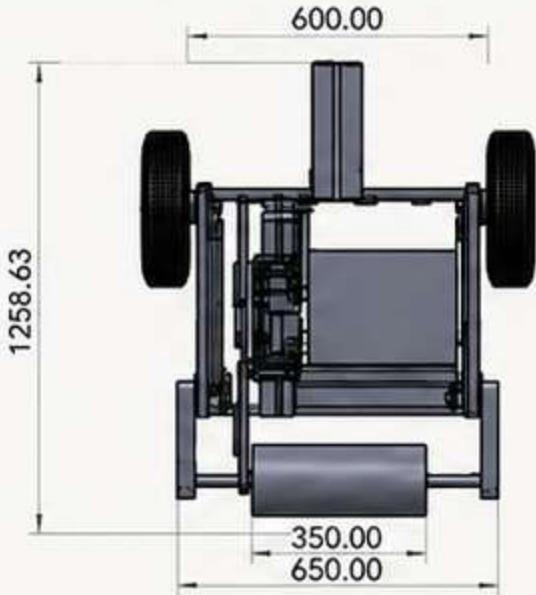


OBJECTIVE
Efficient, compact and
cost-effective solution
for pothole repair

DIMENSIONAL VIEWS (mm)



SIDE VIEW



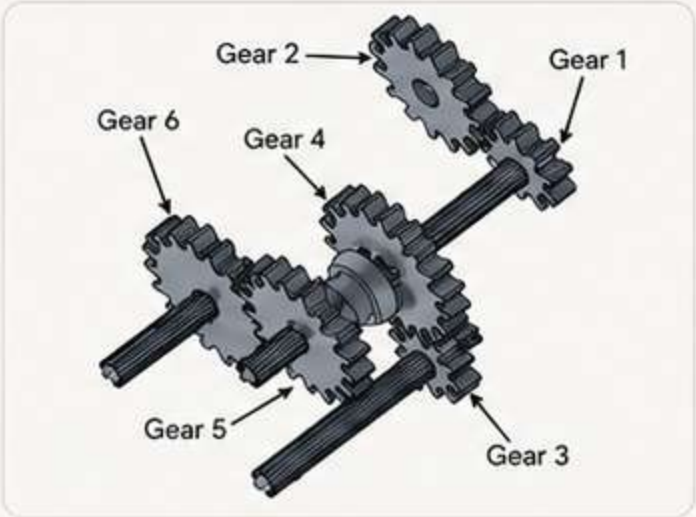
TOP VIEW

Gearbox Systems & Cutter Mechanism

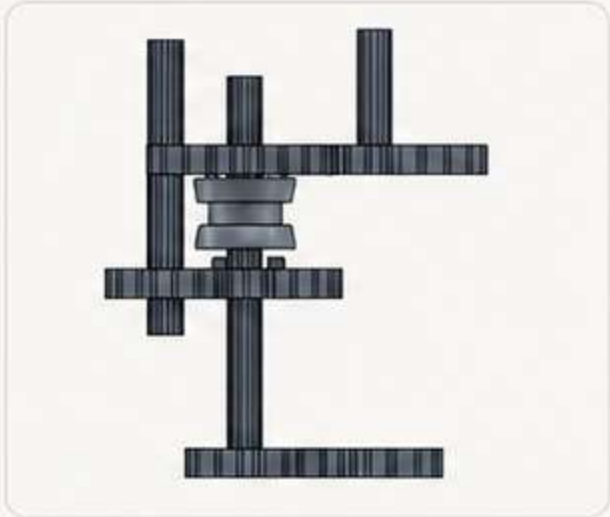
The Road Patcher uses two separate gearboxes. Gearbox 1 drives the cutter and roller for cutting and compaction, while Gearbox 2 drives the wheels for forward and reverse movement.

01 GEARBOX 1 Power Transmission to Cutter and Roller

Gearbox 1 transmits power from the engine to the cutter and roller. It uses multiple gears and a dog clutch to engage either the cutter mechanism or the roller mechanism without stopping the engine.



Isometric View (Gears 1–6)



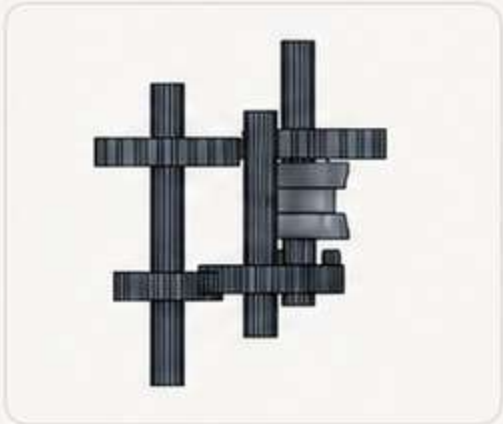
Top View (Gearbox 1)

02 GEARBOX 2 Power Transmission to Wheels

Gearbox 2 is used to drive the wheels for forward and reverse movement. It consists of five gears and allows speed reduction and direction control.



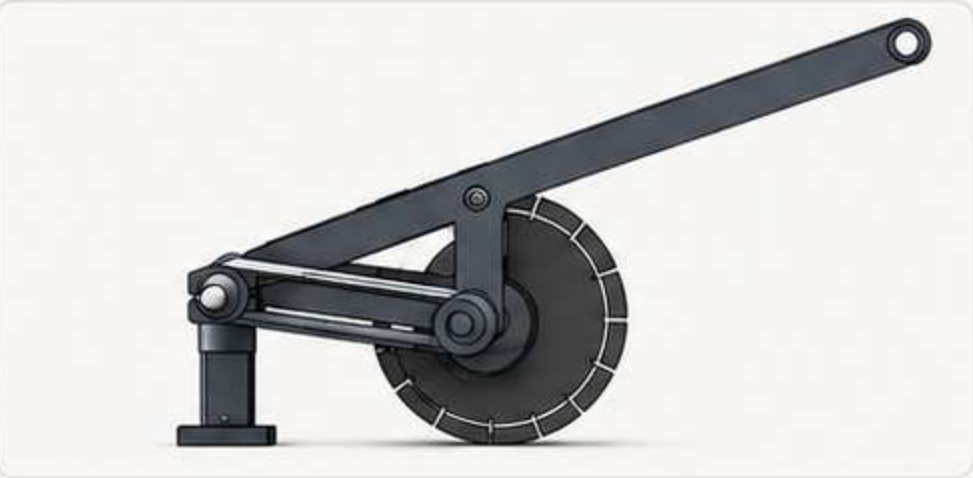
Isometric View (Gears 7–11)



Top View (Gearbox 2)

03 CUTTER MECHANISM

The cutter mechanism is used to cut the pothole area into a uniform shape before filling with asphalt mix. It is powered through Gearbox 1 (cutter output) and operates at 2500 RPM.



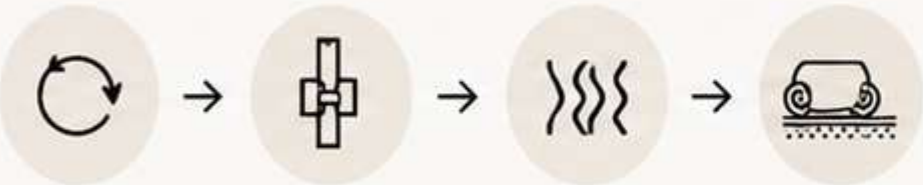
CUTTER SPECIFICATIONS

- Cutter Type : Asphalt Diamond Cutter
- Cutting Diameter : 350 mm
- Operating Speed : 2500 RPM (from Gearbox 1)
- Depth Control : Manual (Adjustable Handle)

Compaction System

The compaction system uses a cylindrical roller to compact the filled asphalt and produce a uniform surface finish. An eccentric mass mechanism inside the roller generates vibration during rotation, improving the compaction effect.

OPERATING PRINCIPLE



Rotational Input

Power from gearbox rotates the roller shaft.

Eccentric Mass Rotation

Offset masses rotate with the shaft, creating imbalance.

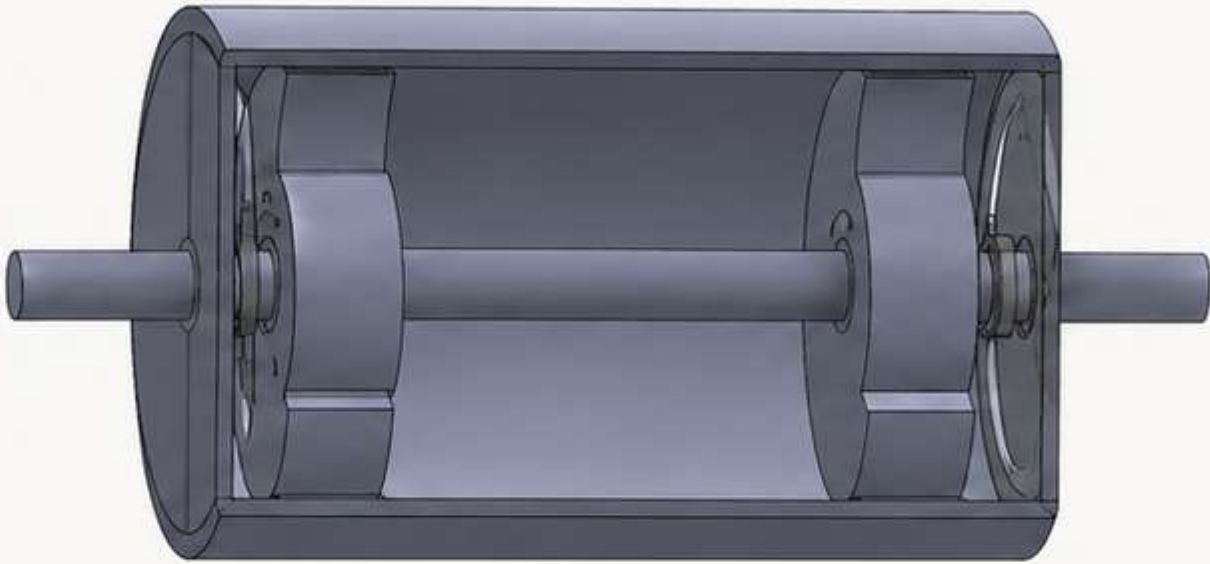
Vibration

Imbalance generates vibrations for effective compaction.

Roller Compaction

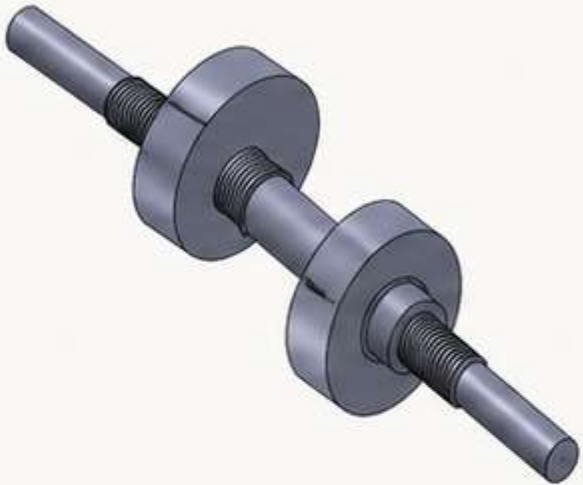
Vibrations compact the asphalt and produce level finish.

VIBRATORY ROLLER (INTERNAL VIEW)



The cylindrical drum houses the eccentric mass assembly on the shaft. As the shaft rotates, the eccentric masses create vibration, ensuring uniform and effective compaction.

ECCENTRIC MASS ASSEMBLY



Two offset masses mounted on the shaft create an unbalanced force during rotation, which produces controlled vibration for improved compaction.



ROLLER TYPE

Single Drum



COMPACTION METHOD

Vibratory



VIBRATION MECHANISM

Eccentric Mass

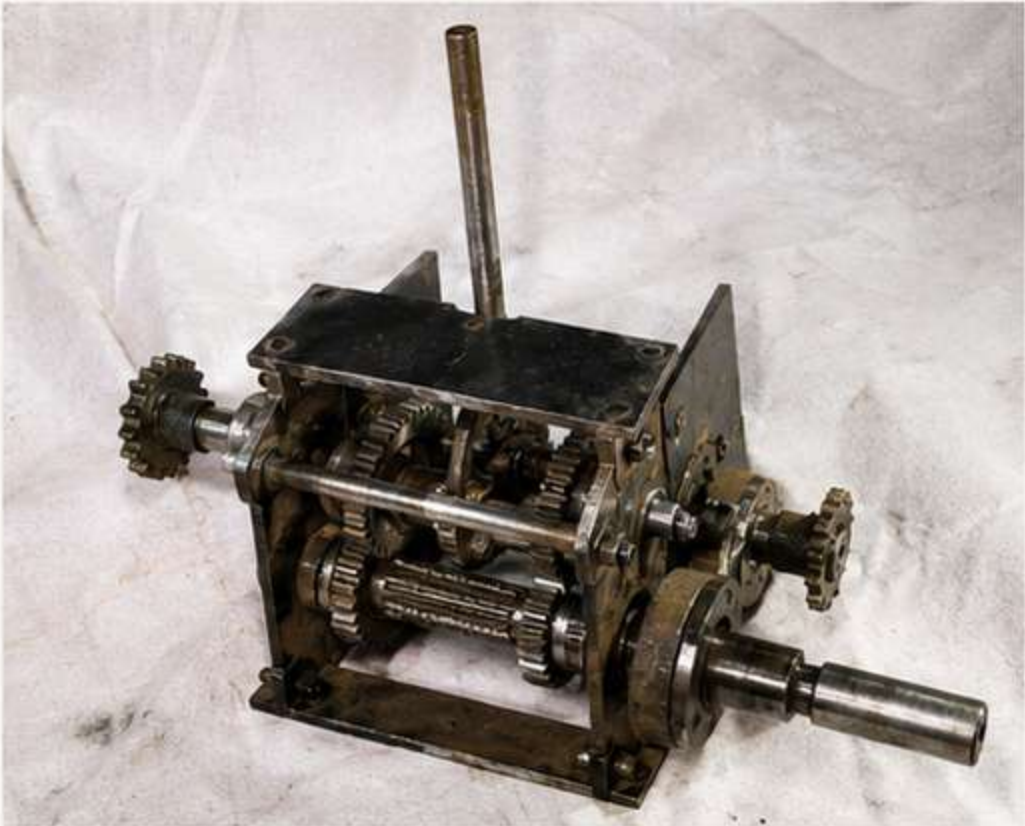


FUNCTION

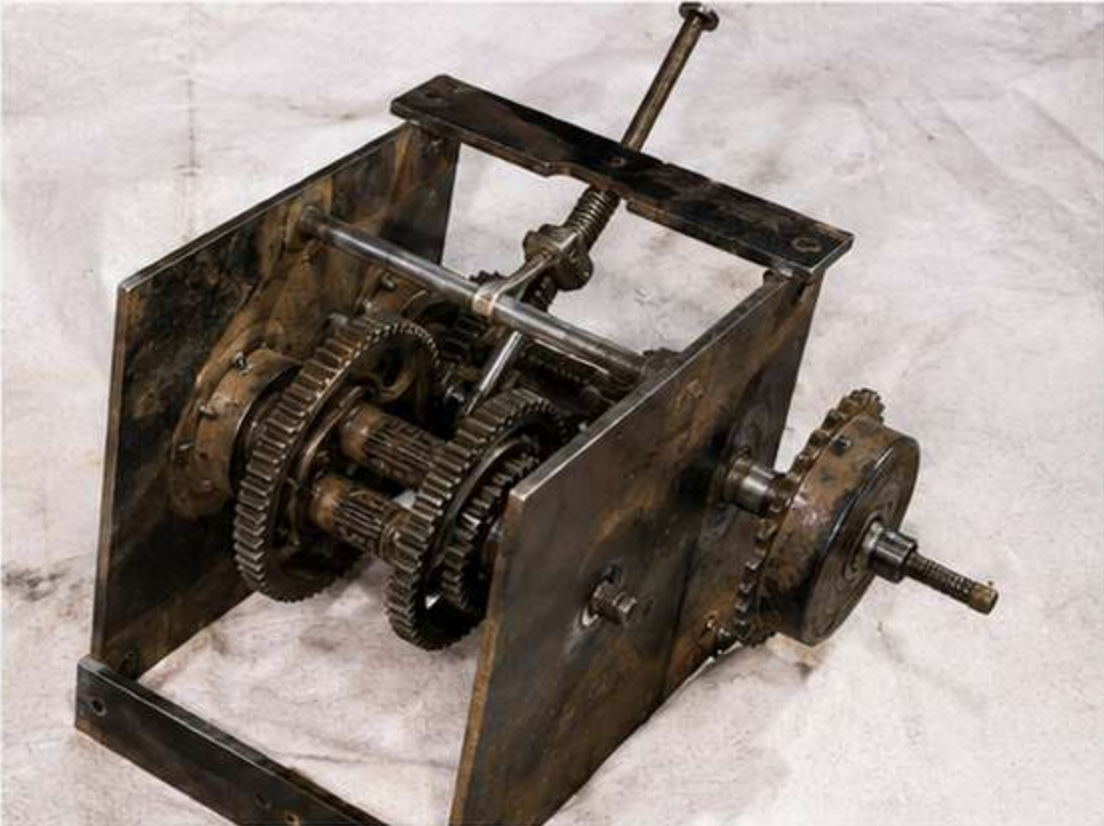
Asphalt Compaction & Surface Finishing

Design to Reality.

The Road Patcher was developed through a systematic process of translating detailed CAD designs into a functional, reliable prototype. Precision fabrication, accurate assembly and rigorous integration ensure that every subsystem works together seamlessly in real road conditions.

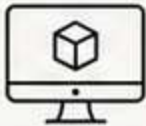


GEARBOX 1 – CUTTER & ROLLER DRIVE
Power transmission to cutter and roller mechanism.



GEARBOX 2 – WHEEL DRIVE (FORWARD & REVERSE)
Power transmission to wheels for forward and reverse movement.

DEVELOPMENT PROCESS



1. CAD DESIGN

3D modelling of components and assemblies using SolidWorks.



2. COMPONENT SELECTION

Selection of engine, gearboxes, rollers, cutter and standard components.



3. FABRICATION

Cutting, welding, machining and drilling to manufacture structural and functional parts.



4. ASSEMBLY & FITMENT

Precise assembly of gearboxes, mechanisms and sub-systems with proper alignment.



5. PROTOTYPE INTEGRATION

Integration of all systems and testing for performance and field readiness.

KEY FOCUS AREAS



DESIGN

- Functional & compact design
- Strength & rigidity analysis
- Ease of operation & maintenance



FABRICATION

- Mild steel structural fabrication
- Precision machining of shafts, gears and keyways
- Strong welds & durable finish



ASSEMBLY

- Accurate alignment
- Proper fitment & lubrication
- Secure fasteners & locking mechanisms



TESTING & INTEGRATION

- Functional testing of all systems
- Load, movement & reverse test
- Field performance validation

Engineering Calculations.

The performance of the Road Patcher depends on proper power transmission, gear reductions and compaction force.

All calculations are based on standard mechanical engineering principles and the actual parameters of the machine components used in this project.

01 CUTTER POWER CALCULATION



INPUT PARAMETERS

- Cutter Diameter (D) : 350 mm (0.35 m)
- No. of Blades (n) : 21
- Cutting Resistance (F) : 100 N
- Angular Velocity (ω) : 261.8 rad/s (2500 RPM)



Power Required (P)
= 4.5 kW ($\approx$ 6.03 HP)

Torque Required (T)
= 17.18 Nm ($\approx$ 17 Nm)

02 MACHINE MOTION POWER



INPUT PARAMETERS

- Machine Mass (m) : 100 kg
- Rolling Resistance Coefficient (μ_r) : 0.02
- Acceleration (a) : 0.14 m/s²
- Velocity (v) : 5 km/h (1.38 m/s)



Total Tractive Force (F)
= 884 N

Power Required (P)
= 1219.92 W
($\approx$ 1.63 HP $\approx$ 2 HP)

03 VIBRATORY ROLLER CALCULATION



INPUT PARAMETERS

- Roller Drum Mass (m) : 16 kg
- Eccentric Mass (m_e) : 2.2 kg
- Eccentric Radius (r) : 3.725 cm (0.03725 m)
- Angular Velocity (ω) : 105 rad/s (1005 RPM)



Static Force (F_{static}) : 157 N

Dynamic Force ($F_{dynamic}$) : 903 N

Total Compaction Force (F_{total}) : 1060 N

Compaction Power : 5.9 kW ($\approx$ 8 HP)



POWER REQUIREMENT SUMMARY

- Cutter Power : $\approx$ 6 HP
- Machine Motion Power : $\approx$ 2 HP
- Roller Compaction Power : $\approx$ 8 HP



DESIGN POWER REQUIREMENT

$\approx$ 10 HP

Based on operating modes, with machine-motion power shared during cutting and compaction operations.



SELECTED ENGINE

150cc Bajaj Petrol Engine

- Maximum Power : 13 HP
- Maximum Torque : 86 Nm (in 1st gear)
- Operating Speed : 6500 RPM

13 HP available vs. $\approx$ 10 HP design requirement.

Materials & Manufacturing.

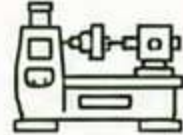
The Road Patcher is fabricated using carefully selected materials and standard manufacturing processes to achieve strength, durability and reliable performance of each mechanism.

Material selection is based on the functional requirements of individual components, considering strength, wear resistance, machinability and cost-effectiveness.

MATERIALS SELECTION			
	FRAME / CHASSIS	Mild Steel (MS Rectangular Pipes)	Provides high strength and stiffness for the main structure while remaining weldable and cost-effective.
	VIBRATORY ROLLER DRUM	Mild Steel (8 mm Thickness)	Offers good strength and wear resistance to withstand compaction loads during operation.
	GEARBOX HOUSING	Mild Steel (MS Plates)	Ensures structural rigidity and durability to protect internal gears and components.
	SHAFTS	Mild Steel	Suitable for transmitting torque and supporting rotating elements.
	GEARS	Hardened Alloy Steel (EN24 / EN36)	Provides high strength and wear resistance for gears.
	CUTTER BLADE	Diamond Tipped Steel (Asphalt Diamond Cutter)	Delivers efficient cutting performance and long edge life on asphalt surfaces.



MANUFACTURING METHODS



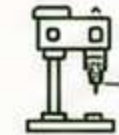
MACHINING

Shafts and gears precision machined using lathe and milling machines.



STICK WELDING (SMAW)

Frame and chassis fabricated using MS pipes and plates and assembled by stick welding.



DRILLING

Accurate drilling operations for mounting components and assemblies.



ASSEMBLY

All components assembled, aligned and fitted to ensure smooth operation.

Project Evaluation.

The Road Patcher was developed as a compact, semi-automatic machine to carry out pothole repair by integrating cutting, filling and compaction in a single process.

This page summarizes the key outcomes of the design, the current limitations of the machine, and the opportunities for future improvement.

DESIGN OUTCOMES



COMPACT & PORTABLE
Compact in size and easy to transport and operate on-site.



INTEGRATED REPAIR PROCESS
Integrates cutting, filling and compaction in a single process.

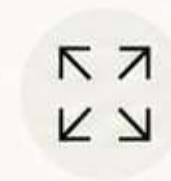


SIMPLE MECHANICAL SYSTEM
Uses gearboxes, chain drive and standard components for reliable operation and maintenance.



VIBRATORY COMPACTION
Single drum vibratory roller is used to compact the filled material.

CURRENT LIMITATIONS



SMALL-MEDIUM REPAIR SCOPE
Intended for small to medium sized pothole repairs.



MANUAL ASPHALT FILLING
Filling of the cut-out section with asphalt mix is done manually.



SURFACE CONDITION DEPENDENCE
Performance varies with uneven surfaces, moisture, debris and other site conditions.



MAINTENANCE REQUIRED
Mechanical transmission and moving parts require regular inspection, lubrication and maintenance.

FUTURE DEVELOPMENT



Improve gearbox design and lubrication for better durability and efficiency.



Add monitoring of engine and vibration parameters for better control.



Use better quality materials and components to improve reliability and life.



Enhance the vibratory system to improve compaction efficiency.

Thank You.

Thank you for viewing my portfolio.

AMALJITH P A

CONTACT



amaljithamalpa@gmail.com



+91 8848370746



linkedin.com/in/amaljithpa